# What Aggregation Does to Epistemic Content: A Behavioral Characterization of Multi-Seat LLM Pipelines

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**Abstract**

Multi-seat pipelines route a question to several specialist models and combine their answers with a final model. We characterize what that final step does to the epistemic content it receives—the labelled assumptions, jurisdictional distinctions, and disclosures of uncertainty the specialists write—by tracking each such marker from the seat that raised it to the finished answer, across roughly 1,400 audited runs on a single consumer machine.

The aggregation step is not a channel. Of the behavior families the specialists raise, it preserves about $72\%$ and discards the rest at a rate that does not depend on how much arrives. It also *invents*: on questions where the specialists raise little, more than a third of what the final answer contains traces to no specialist at all, against a ninth on questions where they raise plenty. Invention is a monotone function of scarcity: pooled across four writing models and $358$ runs, it falls from $0.53$ families when the specialists raise none to zero when they raise all four, while traceability rises from $0\%$ to $100\%$. The step *partially compensates* for thin input rather than conveying it, and a $2.6\times$ distinction the specialists draw between demanding and undemanding questions reaches the page as $1.8\times$. Around that behavior we map what the step responds to and what leaves it unmoved. It responds to its own instructions, gradedly and with per-behavior specificity: output rises with the number of preservation clauses, and each clause moves exactly the behavior it names and no other. It is close to invariant under everything upstream—the volume of qualification the specialists produce (raising it lowers the output, monotonically), the number of specialists, the pipeline topology, and weight-level training at either end under three optimizers and two reward designs. We report these as measured regularities of the aggregation step rather than as an argument for or against building such pipelines.

## 1 Introduction

A multi-seat pipeline is a system with a distinguished component. Several models contribute, but one of them writes the text a user reads, and whatever the others produced reaches the user only through that final act of writing. This paper is about the behavior of that step.

We study it on one kind of content: *epistemic* markers—a number labelled as an assumption rather than a fact, a distinction drawn between two regulatory regimes, a note that some figure may have changed since the model was trained, a calibrated hedge. This content is a good probe for two reasons. It is identifiable in text, so it can be traced from where it was written to where it ends up. And it is the kind of content that matters most when it goes missing: a confident synthesis of hedged inputs misrepresents what the system actually knows.

Our question is descriptive. Given that a specialist wrote a caveat, what happens to it? Given that the specialists collectively wrote more or fewer caveats, what happens to the total? What changes the answer,

Table 1: Provenance of behavior families, per response (648 syntheses).

| Question type | raised | preserved | discarded | invented | traceable |
|---|---|---|---|---|---|
| trigger-heavy | 2.51 | 1.81 | 0.70 | 0.12 | 94% |
| trigger-free | 0.97 | 0.68 | 0.29 | **0.38** | **64%** |

and what does not?

We report three groups of findings. Section 3 follows individual behavior families through the aggregation step and decomposes the output into what was preserved, what was discarded, and what was invented. Section 4 characterizes the one input the step responds to strongly—its own instructions—including how gradedly and how specifically. Section 5 reports the manipulations that leave it essentially unmoved, which is the larger set. Section 7 assembles these into a compact behavioral description.

We are describing one instantiation of a common pattern, not proving a theorem, and Section 8 is explicit about what that licenses.

## 2 Apparatus and measurement

**Pipeline.** A planner decomposes the question and dispatches a self-contained sub-question to each relevant specialist; specialists never see the original query. A final model then writes the answer from their contributions under an instruction that asks it to surface disagreements and to preserve specific things the specialists flagged (Appendix A). Four open-weights models at or below 20B run locally on one 32 GB machine: Phi-4-14B as planner and writer unless stated, with Llama3-Med42-8B, Saul-7B-Instruct and Qwen-Open-Finance-8B as specialists. Variants substitute gpt-oss-20B or Qwen2.5-7B in the writing role.

**What we count.** Four families of epistemic behavior: disclosure that information may post-date training, labelling a quantity as an assumption, distinguishing jurisdictions or regimes, and calibrated hedging. A fifth we defined—distinguishing near-synonymous terms of art—fell below detection on two independent instruments and is dropped. We report both the *density* of markers per thousand characters and, where provenance matters, the number of distinct families present, which is insensitive to repetition.

**Provenance.** Because each specialist's contribution appears verbatim in the writing model's prompt, we can determine for any finished answer which families were *raised* upstream, which of those were *preserved*, and which appear in the output having been raised by nobody. This is the measurement the paper rests on and it requires no additional model calls.

**Instruments.** Pattern matching is primary. Every load-bearing comparison is re-scored by a calibrated entailment model and by blinded pairwise LLM judges shown both texts in each order and required to quote their evidence. Where the three disagree we say so; one finding produced by pattern matching alone did not survive and was retired (Appendix B).

**Scale.** Roughly 1,400 runs, five seeds per condition, bootstrap intervals throughout. Predictions were registered before the runs that test them, including several that failed.

## 3 What the step preserves, discards, and invents

Three regularities appear in Table 1.

**Discard is proportional.** The step drops 28% of raised families on demanding questions and 30% on

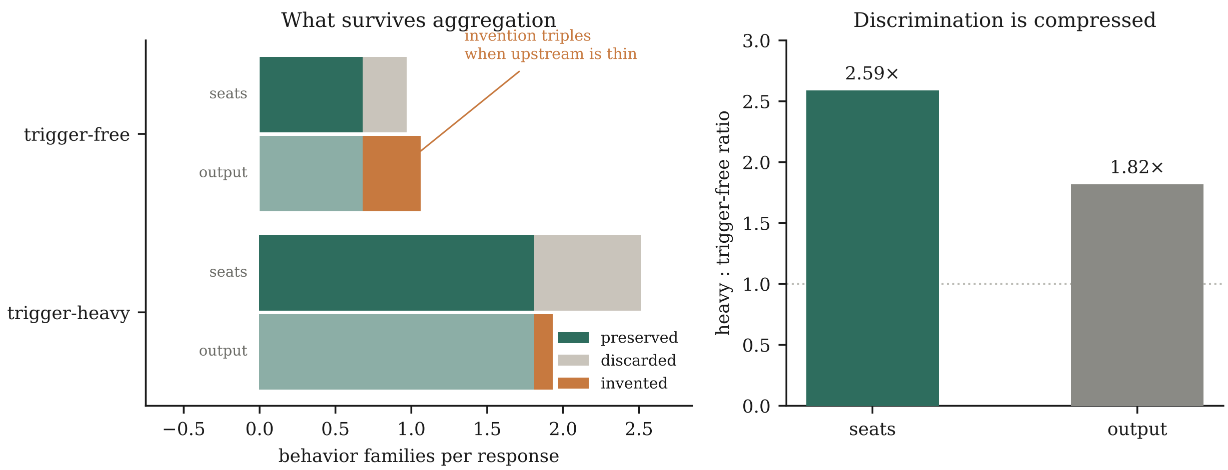


Figure 1: *Left:* for each condition, the upper bar is what the specialists raised, split into preserved and discarded; the lower bar is what the finished answer contains, split into traceable and invented. *Right:* the ratio between demanding and undemanding questions, as the specialists draw it and as it reaches the page. Means over 648 syntheses.

undemanding ones. Loss is close to a constant fraction of what arrives rather than a fixed amount, and does not depend on how much is supplied.

**Invention is not proportional—it compensates.** On demanding questions almost nothing in the output is untraceable ($0.12$ families, $94\%$ traceable). On undemanding questions, where the specialists raise little, invention more than triples in absolute terms and over a third of the output traces to no specialist. The step is not simply lossy; when the supply of genuine qualification is thin, it produces qualification anyway.

**Distinctions compress.** The specialists draw a $2.59\times$ distinction between demanding and undemanding questions ($2.51$ families against $0.97$). What reaches the page is $1.82\times$ ($1.94$ against $1.06$). Loss at the top and invention at the bottom together compress the distinction the upstream models drew. The information that a question warranted little caution is available in the pipeline and is substantially attenuated by the time it is written.

**The controlling variable is supply, not question type.** Pooling 358 runs that carry specialist turns, across four different writing models and nine arms, the relationship is monotone in how much the specialists raised:

| families raised | $n$ | preserved | invented | output | traceable |
|---|---|---|---|---|---|
| 0 | 17 | 0.00 | 0.53 | 0.53 | $0\%$ |
| 1 | 34 | 0.50 | 0.26 | 0.76 | $65\%$ |
| 2 | 82 | 1.06 | 0.29 | 1.35 | $78\%$ |
| 3 | 180 | 1.63 | 0.07 | 1.71 | $96\%$ |
| 4 | 45 | 1.98 | 0.00 | 1.98 | $100\%$ |

Invention falls monotonically as supply rises and reaches zero when the specialists raise all four families; traceability rises from $0\%$ to $100\%$ across the same range ($r = -0.33$ between raised and invented). At zero supply the writer still emits half a family, invented outright.

We describe this as *partial compensation* and deliberately not as a target level. Output does track supply ($r = +0.41$, rising from $0.53$ to $1.98$ across the range); it simply tracks it sub-proportionally, with the

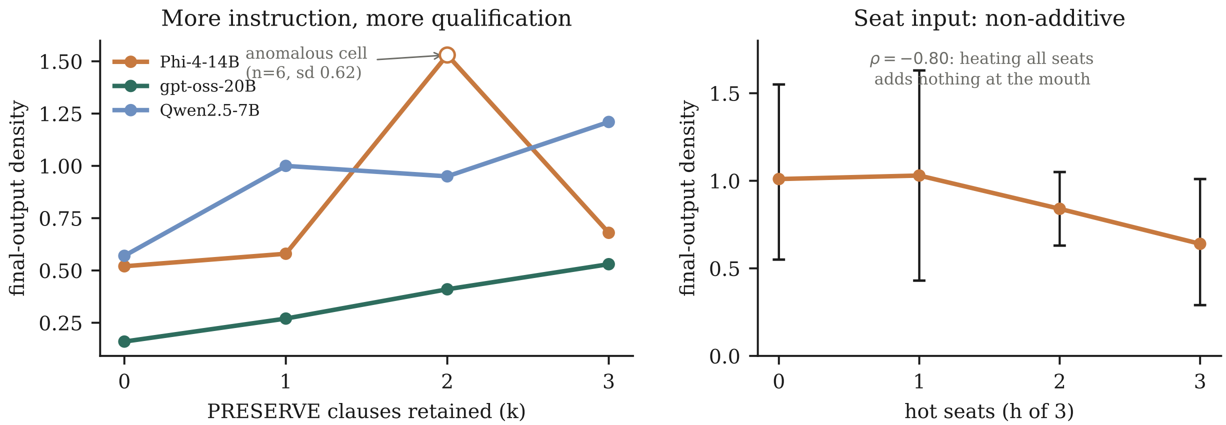


Figure 2: *Left:* output rises with the number of preservation clauses retained in the writing model's instructions, on three different writers. *Right:* giving the same disposition instruction to more specialists leaves the output flat to declining.

shortfall made up by invention that is largest where supply is smallest. A writer with a fixed target would show flat output, and it does not.

This also resolves what looks like a failed replication. One writer shows $100\%$ traceability on undemanding questions, apparently contradicting the pattern—but on those runs the specialists raised $3.00$ families, an oversupply condition, and the writer trimmed ($3.00$ raised, $1.25$ kept) rather than invented. It is the same behavior observed from the other end.

## 4 What the step responds to

One intervention moves the aggregation step substantially: the instructions given to it.

**The response is graded, not switch-like.** Retaining $k \in \{0, 1, 2, 3\}$ preservation clauses raises output on all three writers tested (Spearman $\rho = +1.00, +0.80, +0.80$), spanning 2–3$\times$ on two of them and $1.3\times$ on the third. The relationship is a monotone trend rather than strict monotonicity: only one writer increases at every step, with the other two showing a single-cell dip within the noise of six observations per point. Holding the pipeline, the models, and the specialists' text fixed and changing only these clauses moves output from $0.159$ to $0.525$ on one writer and $0.575$ to $1.211$ on another.

**The response is specific to the behavior named.** Isolating clauses one at a time shows each moving exactly the family it names and leaving the others flat: the numeric-framing clause raises labelled assumptions from $0.003$ to $0.499$; the caveats clause raises cutoff disclosure from $0.000$ to $0.035$; the vocabulary clause raises its own family from $0.000$ to $0.014$. These are targeted specifications rather than general encouragement to be careful, which rules out the alternative that instructions simply put the writer in a more cautious mood.

**Aggregate movement goes to whichever clause matches the writer's profile.** Under one writer, the numeric clause alone accounts for $126\%$ of the full block's effect while the others land on baseline—not because they fail, but because the behaviors they name are ones this writer barely produces in the first place. Which behavior dominates differs by writer: cutoff disclosure accounts for most of the composite under two writers ($0.554$, $0.909$) and almost none under a third ($0.018$). Any recommendation about which clause to write is therefore conditional on the writer.

**Clauses interfere mildly.** Single-clause effects sum to $+0.41$ against the assembled block's $+0.35$, and the strongest single clause alone slightly exceeds the full block. Adding clauses aimed at behaviors a writer does not produce appears to compete with the one that works.

## 5 What leaves the step unmoved

The larger set of findings is negative in form and, we think, more informative for it: these are the invariances that constrain what a behavioral account can look like.

**Invariant under upstream volume—and inversely responsive.** Raising what the specialists produce does not raise the output. Prompting a specialist to qualify more doubles what it writes ($0.84 \rightarrow 1.82$) while the finished answer falls ($1.21 \rightarrow 0.92$); training it on qualification-rich examples reproduces this ($1.75$ at the seat, $0.92$ at the mouth). Extending the treatment to more specialists lowers the output monotonically, from $1.01$ at none to $0.64$ at all three ($\rho = -0.80$). Over-dense input does not merely fail to transmit; on one writer it inverts, $1.17 \rightarrow 0.61$.

**Invariant under the number of demands.** Constructing questions that demand one, two, three, or four distinct kinds of qualification, with length held constant, leaves the output flat ($\rho = +0.20$, non-monotone, $L$=4 overlapping $L$=1). Across the same range the writer engages 1.2–1.7 families regardless of how many were asked for. Between-question variance at fixed demand exceeds any variation across demand.

**Invariant under topology.** Holding the model fixed in every role and varying only the arrangement, three specialists merged plainly reach $0.16$ against a single model's $0.14$; two full rounds of debate reach $0.16$; a single model that drafts, critiques itself and revises reaches $0.08$. A single model given a good instruction and no specialists at all reaches $0.50$, against the full pipeline's $0.57$, intervals overlapping.

**Invariant under weight-level training, at both ends.** Reference-free preference optimization on a specialist installs no additional qualification ($0.85$ against $0.84$ untrained) and tripling the training data changes nothing, though the larger corpus raised the model's own accuracy at distinguishing good examples from bad from $0.48$ to $0.94$. Replication on a second specialist of a different lineage produced no effect; on a third, an effect registered in advance failed to appear. Training the *writing* model directly leaves its output where it started ($0.89$ against $1.03$), and so does an on-policy retry whose reward was computed on the finished answer and shaped to reward qualification where warranted and penalize it where not ($0.60$ against $0.56$). Notably, after training the writer on precisely the behavior its instructions elicit, those instructions still worked at full strength ($1.58\times$ against an untrained $1.82\times$), which suggests the two act on different things.

**Not invariant under choice of writer.** Substituting the model in the writing role changes the output under identical input, which is the one structural change that does move it.

## 6 The one output with no single-model analogue

Everything measured so far concerns how much epistemic content the step emits, and on that axis a well-instructed single model matches the pipeline. There is one output where no such comparison is possible.

The writing instruction asks, before the answer is drafted, for an explicit enumeration of tensions between the specialists—places where following one contribution makes another harder. This is the only part of the pipeline's output that a single model cannot produce by construction: identifying a conflict requires more than one source to conflict.

The separation is categorical rather than a matter of degree (Table 2). Arrangements that pass specialist text

Table 2: Tension enumeration by arrangement. Only arrangements with a genuine aggregation step produce one.

| Arrangement | runs | responses with a tensions section | tensions per response |
|---|---|---|---|
| council (all variants) | 195 | 86–100% | 3.0–3.9 |
| flat merge | 50 | 0% | 0.00 |
| single model + instruction | 90 | 0% | 0.00 |

to a writer without asking for tensions produce none at all; the council produces close to four per response, reliably.

**Are they grounded?** Given the invention behavior of §3, the obvious concern is that these conflicts are manufactured. Many tensions cite a specific figure and attribute it to a named specialist, which makes them checkable: the figure either appears in that specialist's contribution or it does not. Across 210 such tensions, **89%** cite a figure that is genuinely present upstream (median 100% per response); 11% cite one that appears nowhere.

This is a notably better rate than the step's handling of epistemic content on thin questions, where 36% of what it emits is untraceable. The step invents caveats far more readily than it invents disagreements—which are different behaviors, and the asymmetry is the more interesting result.

**An extension that failed, and why it is worth reporting.** The 89% covers only tensions citing a figure. We attempted to extend the check to the 139 qualitative ones using the entailment instrument that validates our other measurements, and it does not transfer. On the figure-citing subset where both methods apply, entailment scores 16% grounded against figure-matching's 89%, and the two agree on only 23% of items. Decomposing each tension into its component claims—a tension is typically a compound contrastive statement, "A recommends $x$, but B assumes $y$"—raises the entailment rate to 48% with at least one clause supported, and the median best-clause entailment sits at 0.57, just under our 0.60 threshold.

The diagnosis is that entailment is the wrong relation here. A tension clause is a *paraphrased summary* of what a specialist wrote, not a restatement of it, and no single upstream sentence entails a claim that combines two contributions with a conflict framing. What the instrument measures on this material is paraphrase distance rather than groundedness, which is why its scores cluster at the decision boundary. We therefore report the figure-based result, which is unambiguous, and record the qualitative majority as not measurable by the instruments we have. Establishing it would need human judgement or a judge shown both contributions and asked whether the conflict is real.

This is the third occasion in this work on which an instrument disagreed with another and the disagreement, rather than either number, was the finding.

## 7 A behavioral description

The regularities above are compactly summarized as follows. The aggregation step behaves like a writer with a target level of epistemic content rather than a channel that conveys it. That target is set by the step's instructions, gradedly and per-behavior, and modulated by which model occupies the role. Content arriving from upstream is preserved at a roughly constant rate when it is plentiful and supplemented by invention when it is scarce, with the consequence that distinctions drawn upstream reach the output attenuated.

Two predictions follow that we did not set out to test but that the data support. First, interventions that increase upstream qualification should be counterproductive, because they push past the target and the excess is trimmed—which is what the seat-tuning and multiple-specialist results show. Second, a system's apparent indifference to input should be strongest where input is thinnest, because that is where invention is doing the most work— which is what the $94\%$ against $64\%$ traceability split shows.

The description also has a boundary. It concerns epistemic content that passes *through* the step. The tension enumeration of §6 is content the step *constructs*, and it behaves differently: it is far more faithful to upstream ($89\%$ of checkable tensions grounded) than the step's handling of caveats on thin input ($64\%$ traceable). Whatever produces compensating invention does not operate on constructed conflicts to the same degree.

One thing this description does not explain is why instructions move the step when weight-level training does not. Three accounts are consistent with our data—that the capability is present and instructions retrieve rather than install it; that the target behavior is conditional on information available only at inference; and that training and instructing act on different mechanisms, as the surviving instruction effect after training suggests—but we cannot distinguish among them, and our training was small enough that scale remains a live alternative.

## 8 Limitations

Of five behavior families we set out to track, one fell below detection on both instruments, so we report four; the instruction effect is clear on the combined measure but not on any single behavior alone. Pattern matching misses paraphrase and rewards brevity, and although two further instruments agree with it on the load-bearing comparisons, all three are automatic and none is anchored to human judgement.

Everything ran at or below 20B parameters on one consumer machine, with five seeds and seven questions we wrote ourselves plus eleven added for specific tests. A frontier model in the writing role could behave differently. The topology comparison uses one model family throughout and one implementation of each arrangement. Full policy-gradient reinforcement learning could not be attempted: on a 34 GB machine, group-relative optimization of a 7B writer with $2.2$k-token prompts and a $152$k vocabulary exhausts memory before the first update, so the invariance under training is established for offline preference optimization and an on-policy best-of-$n$ surrogate only.

We report defects in our own record rather than repairing them. One model intended for a lineage comparison produced degenerate output—26 of 30 answers were fragments ending mid-sentence—and that arm is withdrawn, taking the comparison with it. A claim about the pipeline's behavior on undemanding questions was found, after publication of a figure computed from it, to have been measured on runs where the planner consulted no specialists and the writing instruction was consequently never applied; those runs were relabelled and the comparison rerun on questions the planner does engage with. In our earliest arms, five runs per question are one run from an earlier session plus four from a later one, and 133 runs record a batch label where a model identifier belongs.

## 9 Implications

For practitioners the description above has direct consequences. Effort spent making upstream components more epistemically careful is not merely wasted but counterproductive at the output. Effort spent on the writing model's instructions is the intervention with measured, graded effect, and should be written for the behaviors that writer actually produces. Systems should be evaluated on questions that warrant no qualification as well as those that do, since the invention behavior is invisible otherwise.

For the study of such systems, the finding we would most like to see tested elsewhere is compensating invention. If it generalizes, then a pipeline's apparent robustness—output that changes little as inputs vary—

is partly an artifact of the aggregation step filling gaps, and evaluations that measure only what the output contains, without asking where it came from, will systematically overstate how much of it is grounded.

**Reproducibility.** Code, prompts, all runs, pre-registrations, protocol amendments, refuted predictions, and integrity audits: `https://github.com/srbobo/council-of-experts-research`.

## A The writing model's instruction

Two steps: identify tensions between the specialists, then write the integrated answer. The second step carries five requirements, of which three are preservation clauses. Verbatim:

> **1.** Acknowledge the tensions you just identified—not abstractly, but at the points where they bite the answer.
>
> **2. PRESERVE numeric framing**—if a specialist labeled a number as a modeled assumption, your synthesis MUST also label it as an assumption (use "modeled at," "assumed"). Never adopt a specialist's modeled number as a fact.
>
> **3. PRESERVE precise vocabulary**—if a specialist used a specific legal term, jurisdiction, FDA pathway, or clinical concept, preserve it; do not smooth out precision in the name of accessibility.
>
> **4. PRESERVE caveats**—if a specialist flagged training-cutoff uncertainty or data fragility, propagate that flag into your synthesis.
>
> **5.** Use whatever structure best fits the user's question.

Clauses 2–4 are conditional on something a specialist did; clause 1 is not, and tested alone it changes nothing. Note that the block specifies what to preserve and never what not to add, which is consistent with the invention behavior of §3.

## B Instruments

Pattern matching over four families is primary. An entailment model calibrated against construction-labelled pairs and blinded pairwise LLM judges, shown both texts in each order and required to quote evidence, check every load-bearing comparison. One finding did not survive: an apparent *increase* in hedging on one specialist appeared under pattern matching alone and neither other instrument saw it. We retired it as an artifact of the measure.